# Intrinsic Finite Element Methods for Fourth-Order Elliptic PDEs on Surfaces

Hao Zhou

School of Mathematical Sciences, Peking University

Advisor: Shuonan Wu

CSIAM 2026 · 6.28 · Chengdu

# Outline

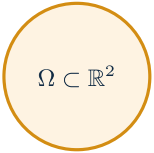
# What is a Surface PDE?

**Surface PDEs:** unknowns are defined on a curved space.

closed surface $\partial\Gamma = \emptyset$

planar curved domain $\Omega \subset \mathbb{R}^2$

closed surface $\Gamma$




curved-boundary plane

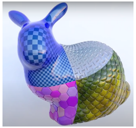
## Problems constrained to an interface

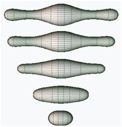
Membranes phase fields on biological interfaces.

Graphics texture synthesis, mesh processing, visual effects.

Thin shells elastic shells, large bending.

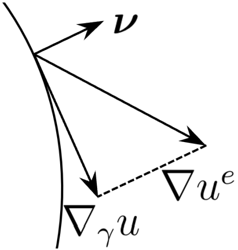
Geometric flows smoothing, shape optimization.





# What is a Surface PDE?

**Surface PDEs:** unknowns are defined on a curved space.




Extrinsic: projection from $\mathbb{R}^3$

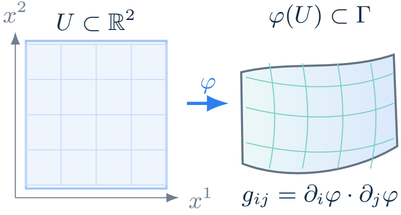
$$\mathbf{P} = I - \nu \otimes \nu, \qquad \nabla_\Gamma u = \mathbf{P} \nabla u^e, \qquad \operatorname{div}_\Gamma X = \operatorname{tr}\big(\mathbf{P} \nabla X^e\big).$$

Intrinsic I: exterior calculus

$$\nabla_\Gamma u = (du)^\sharp, \qquad \operatorname{div}_\Gamma X = -\delta(X^\flat).$$

Intrinsic II: metric

$$\nabla_\Gamma u = g^{ij} \partial_j u \, \partial_i, \qquad \operatorname{div}_\Gamma X = \frac{1}{\sqrt{|g|}} \partial_i \Big( \sqrt{|g|} \, X^i \Big) .$$



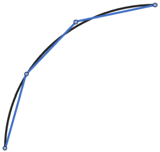

Laplace–Beltrami problem $\Gamma$ closed, $C^2$

$$-\Delta_\Gamma u := -\operatorname{div}_\Gamma \nabla_\Gamma u = f, \qquad u \in \mathring{H}^2(\Gamma),$$

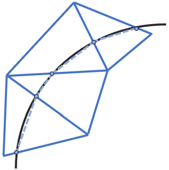
$$(\nabla_\Gamma u, \nabla_\Gamma v)_\Gamma = (f, v)_\Gamma \qquad \forall v \in \mathring{H}^1(\Gamma).$$

# The first step of discretization

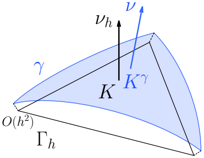
## Approximate surfaces



Isoparametric FEM
Surface FEM



Cut FEM
Trace FEM



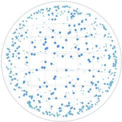

## Surfaces data

explicit $X(\theta, \varphi)$ for $S^2$

metric $ds^2 = d\theta^2 + \sin^2\theta \, d\varphi^2$

level set $\Phi(x) = |x|^2 - 1$

point cloud $\{x_i\}_{i=1}^N \subset S^2$



### Pull everything back to a planar PDE?

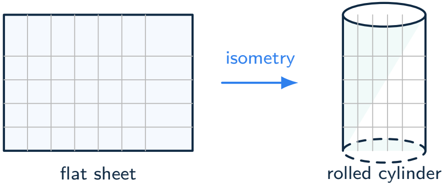
- Need atlas, partition of unity.
- Quadrature issues.

### Use Trace FEM?

- Trace FEM is excellent for moving surfaces.
- But we seek an intrinsic discretization.

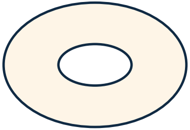
# What is an intrinsic discretization?

**Geometric criterion**
isometry invariant: not depend on embedding




**Topological criterion**
holes, cohomology FEEC

**Analytic criterion**
lowest surface regularity assumption

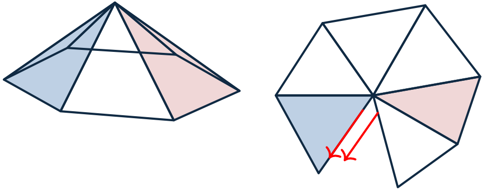


**Geometry on piecewise linear surface**
no $C^0$ tangent vector field

- Extrinsic: elements are tangent to different planes
- Intrinsic: obstructed by the angle deficit

$\Rightarrow$ no $C^1$ scalar field

DDG DEC



# Fourth-order elliptic PDEs on surfaces

Find $u \in \mathring{H}^2(\Gamma)$ such that

- surface biharmonic: $\Delta_\Gamma := \mathrm{div}_\Gamma \nabla_\Gamma$

$$(\Delta_\Gamma u, \Delta_\Gamma v) = (f, v).$$

- surface Kirchhoff plate: $\nabla_\Gamma^2 = \boldsymbol{\nabla}_\Gamma \nabla_\Gamma$

$$(\nabla_\Gamma^2 u, \nabla_\Gamma^2 v) = (f, v).$$

- surface Stokes stream function form: $\mathrm{H}_\Gamma := \mathrm{E}_\Gamma(\mathrm{curl}_\Gamma)$

$$(\mathrm{H}_\Gamma u, \mathrm{H}_\Gamma v) = (f, v).$$

Ignoring lower-order terms.

## Unified formulation

$$a(u, v) = (\Delta_\Gamma u, \Delta_\Gamma v) - \alpha(K \nabla_\Gamma u, \nabla_\Gamma v).$$

| Problem | $\alpha$ |
|---|---|
| Biharmonic | 0 |
| Kirchhoff plate | 1 |
| Surface Stokes | 2 |

**Goal**: A unified discretization without explicit discrete curvature $K_h$.

- avoid approximation errors
- generality as planar case
- curvature treated implicitly

Distributional FEM Regge metric

# Existing methods

Surface FEM only

| **Reference** | **Method** | **Isom. inv.** | **Free topol.** | **Generic no $K_h$** | **Optimal conv.** | **No param.** |
|---|---|---|---|---|---|---|
| [larsson2017] [neilan2024] | 4th-order primal: $C^0$-IP | ✓ | ✓ | ✓ | ✓ | × |
| [cai2024] | 4th-order primal: reconstruction | × | ✓ | ✓ | △ | × |
| [wu2025] | 4th-order primal: NZT, Piola dof | × | ✓ | ✓ | ✓ | ✓ |
| [wu2026] | 4th-order primal: stabilized Morley | ✓ | ✓ | ✓ | ✓ | ✓ |
| [reusken2020] | 4th-order mixed: splitting | ✓ | × | × | ✓ | ✓ |
| [walker2022] | 4th-order mixed: HHJ | ✓ | ✓ | × | △ | ✓ |
| [wu2026] | 4th-order mixed: decoupled | ✓ | × | ✓ | ✓ | ✓ |
| [bonito2020] | Stokes: div-conforming | ✓ | ✓ | ✓ | ✓ | × |
| [hardering2024] | Stokes: penalty normal | × | ✓ | ✓ | ✓ | × |
| [demlow2024] [demlow2025] | Stokes: Piola dof | × | ✓ | ✓ | ✓ | ✓ |
| [nochetto2025] | Stokes: without inf-sup | ✓ | × | × | ✓ | ✓ |

✓: yes; ×: no; △: partially / formulation-dependent;

# Surface nonconforming FEMs

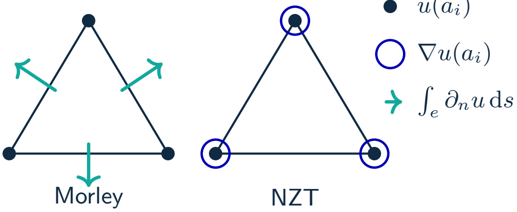
## Surface Morley FEM

- Same FE space as planar case
-
$$a_h(w,v) := (X_{\Gamma_h} w,\, X_{\Gamma_h} v)_{\Gamma_h} + \sum_{e\in\mathcal{E}_h} h_e^{-3} \int_e [w]\,[v]\,\mathrm{d}s_h.$$

## Surface New Zienkiewicz-Type FEM

- Nonstandard vector-valued DoFs
- Stronger continuity
  $\Rightarrow$ No stabilization required in tests

## Degree of freedoms




| **Method** | **Isom. inv.** | **Free topol.** | **Generic no $K_h$** | **Optimal conv.** | **No param.** |
|---|---|---|---|---|---|
| Morley | ✓ | ✓ | ✓ | ✓ | ✓ |
| NZT | × | ✓ | ✓ | ✓ | ✓ |

# Surface nonconforming FEMs

Numerical results for the sNZT element

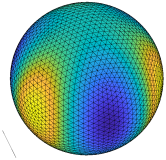


Figure: Numerical approximation of $u = r^{-3}(3x^2y - y^3)$ on the unit sphere.

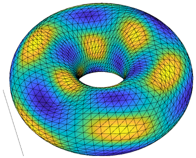


Figure: Numerical approximation of $u = \sin(3\phi)\cos(3\theta + \phi)$ on a torus.

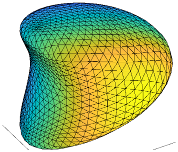


Figure: Numerical approximation of $u = y$ on an implicitly defined surface.

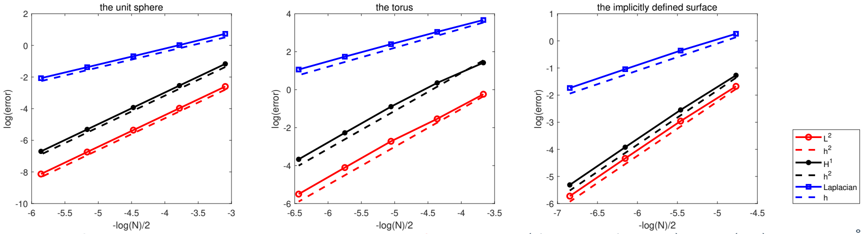



Figure: Error for the surface NZT element using the stabilization-free scheme $(\Delta_{\Gamma_h} u_h, \Delta_{\Gamma_h} v_h) = l_h(v_h) \quad \forall v_h \in \mathring{V}_h$, in the three test cases.

# Surface mixed FEMs

**Problems**:

$$(\Delta_\Gamma u, \Delta_\Gamma v) - \alpha(K\nabla_\Gamma u, \nabla_\Gamma v) = (f, v).$$

**Splitting**: $\phi = \Delta_\Gamma u$ (Reusken 2018, Elliott 2019)

$$\begin{aligned}(\phi, v) + (\nabla_\Gamma u, \nabla_\Gamma v) &= 0,\\ (\nabla_\Gamma \phi, \nabla_\Gamma \eta) + \alpha(K\nabla_\Gamma u, \nabla_\Gamma \eta) &= -(f, \eta).\end{aligned}$$

**HHJ**: $\sigma = \nabla_\Gamma^2 u$ (Walker 2022)

$$\begin{aligned}(\sigma, \tau) + b(\tau, u) &= 0,\\ b(\sigma, v) + (\alpha - 1)(K\nabla_\Gamma u, \nabla_\Gamma v) &= (f, v).\end{aligned}$$

$$\boxed{\text{second-order variable}}$$

$$\Downarrow$$

$K$-term

$$\Downarrow$$

$$\boxed{\text{geometric error / no general}}$$

**Idea**: using first-order auxiliary variable

$$\boxed{\nabla_\Gamma u}$$

# Surface decoupled schemes

**Hodge decomposition**: For $\boldsymbol{g} \in \boldsymbol{L}_t^2(\Gamma)$,

$$\boldsymbol{g} = \nabla_\Gamma \psi + \operatorname{curl}_\Gamma \phi + \boldsymbol{\xi}, \qquad \psi, \phi \in \mathring{H}^1(\Gamma), \quad \boldsymbol{\xi} \in \mathcal{H}_\Gamma.$$

$\mathcal{H}_\Gamma$ is hard to handle $\Rightarrow$ topological restriction (simple connected)

**Decoupling method**

$$\begin{aligned} H^2\text{-problem} \quad &\rightsquigarrow \quad H^1 \ + \ \text{Stokes} \ + \ H^1 \\ H^m\text{-problem} \quad &\rightsquigarrow \quad \underbrace{H^1 + \cdots + H^1}_{m-1} + \text{Stokes} + \underbrace{H^1 + \cdots + H^1}_{m-1}. \end{aligned}$$

**Meanings**

- Bridge between Stokes and fourth-order problems
- For planar curved domains: boundary DoFs improve accuracy
- Naturally extends to higher-order elliptic problems

# Conclusion

### Take-away

- Intrinsic discretization: embedding-free, topology-independent
- An intrinsic index: unified treatment of fourth-order problems
- Primal: stabilized nonconforming FEM; Mixed: decoupled formulation

### Ongoing works

- VEM, HDG, tensor finite elements on surfaces
- SFEM for geometric quantities
- Surface viscoelasticity problems

### Related papers

[1] S. Wu and H. Zhou. *A Stabilized Nonconforming Finite Element Method for the Surface Biharmonic Problem*. SINUM, 2025.
[2] *Stabilized Morley FEM for Surface Stokes in Stream-Function Form: Optimal Convergence via a New Geometric Estimate*. In preparation.
[3] *A Decoupled Finite Element Method for the surface Kirchhoff Plate Problem*. In preparation.

# THANK YOU!

zhouhao23@pku.edu.cn
https://hao23zh.github.io